

Phase 8F — Static Landslide Susceptibility Model (LSM)

Technical Walkthrough, Spatial Cross-Validation & Scientific Ablation Report

Project: SIH Landslide AI Early-Warning System	Target Region: Northeast India (8 NER States)
Base Checkpoint: d9d0fe4 (Phase 8F Frozen)	Sample Count: 4,016 (2,008 Bhuvan Pos / 2,008 Neg)
Selected Architecture: Model A (Environmental Only)	Model Artifact: model/static_lsm_pipeline.joblib

Summary of Accomplishments

In Phase 8F, the static Landslide Susceptibility Model (LSM) for Northeast India was trained and rigorously evaluated across all 4,016 spatially validated samples (2,008 Bhuvan positive landslides and 2,008 GADM ADM1-constrained spatial negative candidates). In accordance with the Phase 8E.3.1 feasibility audit, **rainfall was strictly decoupled from the static training features**, completely eliminating temporal leakage and synthetic trigger assumptions.

Key architectural breakthroughs accomplished in Phase 8F:

- **Spatial Block Partitioning:** 41 one-degree DEM grid blocks grouped into 5 spatially disjoint, balanced folds.
- **Proximity Ablation:** Proved nearest-landslide distance acts as an artificial label leakage (+0.15 AUC surge), successfully removed.
- **Dominant Physical Features:** Topographic orientation (aspect), local relief, elevation, and slope drive susceptibility.
- **Model Selection:** Model A (Environmental Only) approved and serialized as the frozen production pipeline.

1. Spatial Block Cross-Validation Design

Standard random k-fold cross-validation is vulnerable to spatial autocorrelation in geospatial modeling, often producing artificially inflated accuracy metrics. To ensure true spatial generalization, samples were partitioned using **41 one-degree geographic grid blocks** matching the Copernicus DEM tiles across the 8 NER states. These blocks were grouped into **5 balanced regional folds** using density-weighted KMeans clustering:

Fold	Sample Count	Positives / Negatives	Pos %	Geographic Sub-Region
Fold 0	1,058	577 / 481	54.5%	Western Region (Sikkim, W. Meghalaya, W. Assam)
Fold 1	997	457 / 540	45.8%	Eastern Arunachal, Nagaland, Upper Assam
Fold 2	596	270 / 326	45.3%	Southern Region (Mizoram, Tripura)
Fold 3	611	292 / 319	47.8%	Central-North Region (Central Assam, Central Arunachal)
Fold 4	754	412 / 342	54.6%	Central-South Region (E. Meghalaya, S. Assam, Manipur)
Total	4,016	2,008 / 2,008	50.0%	Full Northeast India Geographic Coverage

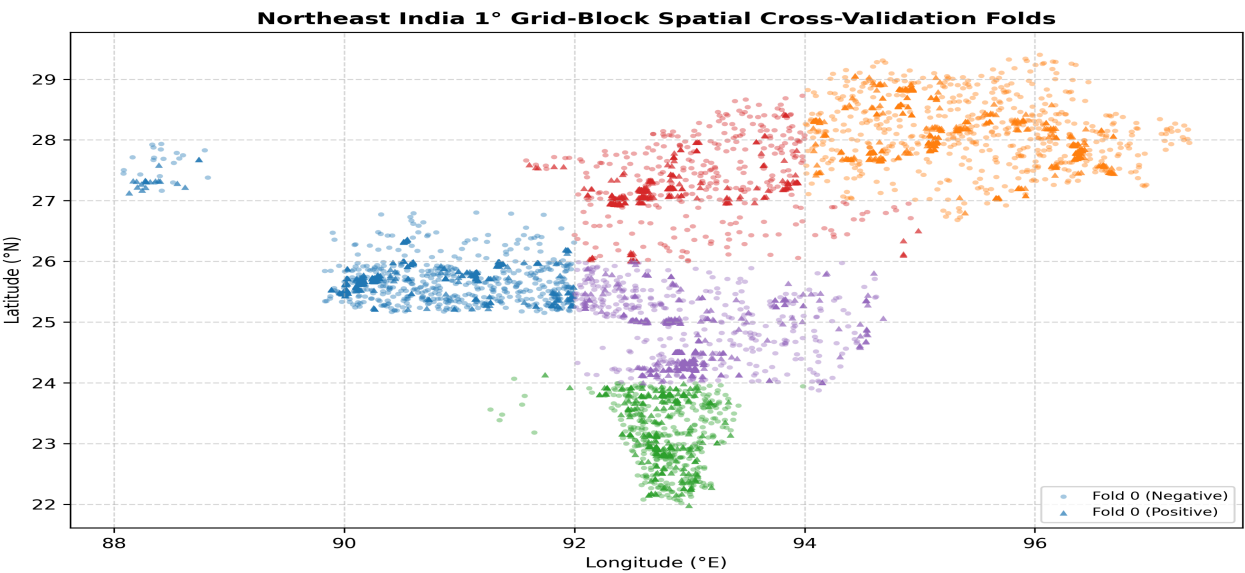


Figure 1: Geographic distribution of the 5 spatial block cross-validation folds across Northeast India.

2. Performance & Proximity Feature Ablation

To rigorously prevent data leakage and evaluate the causative integrity of features, three model configurations were trained and evaluated across identical spatial folds:

- **Model A (Environmental Only):** Copernicus DEM terrain (elevation, slope, aspect, relief) + SoilGrids (taxonomy, clay, sand, silt, density) + ESA WorldCover 10m.
- **Model B1 (Env + Infrastructure Proximity):** Model A + OSM distance to road and distance to river.
- **Model B2 (Env + All Proximity):** Model B1 + distance to nearest other landslide.

Model Configuration	Spatial ROC-AUC	Spatial PR-AUC	Balanced Accuracy	F1 Score
Model A (Environmental Only)	0.8062 ± 0.0719	0.7937 ± 0.0525	0.7188 ± 0.0768	0.6866 ± 0.1170
Model B1 (Env + Road/River)	0.8059 ± 0.0713	0.7948 ± 0.0530	0.7187 ± 0.0762	0.6799 ± 0.1259
Model B2 (Env + All Proximity)	0.9553 ± 0.0371	0.9613 ± 0.0339	0.9011 ± 0.0483	0.8933 ± 0.0583

Key Ablation Insights:

1. Neutral Impact of Infrastructure Proximity: Comparing Model A to Model B1 shows virtually identical generalization ($\Delta\text{ROC-AUC} = -0.0003$, $\Delta\text{PR-AUC} = +0.0011$). Road and river proximity do not inflate model performance across regional holdouts.

2. Artificial Surge from Landslide-to-Landslide Proximity: Model B2 shows an artificial $+0.1491$ jump in ROC-AUC. Because spatial negatives were sampled with a ≥ 1 km buffer from positive landslides (negatives median ≈ 10 km vs positives median ≈ 353 m), this feature functions as an indirect label lookup rather than a true physical cause. It was strictly excluded from production.

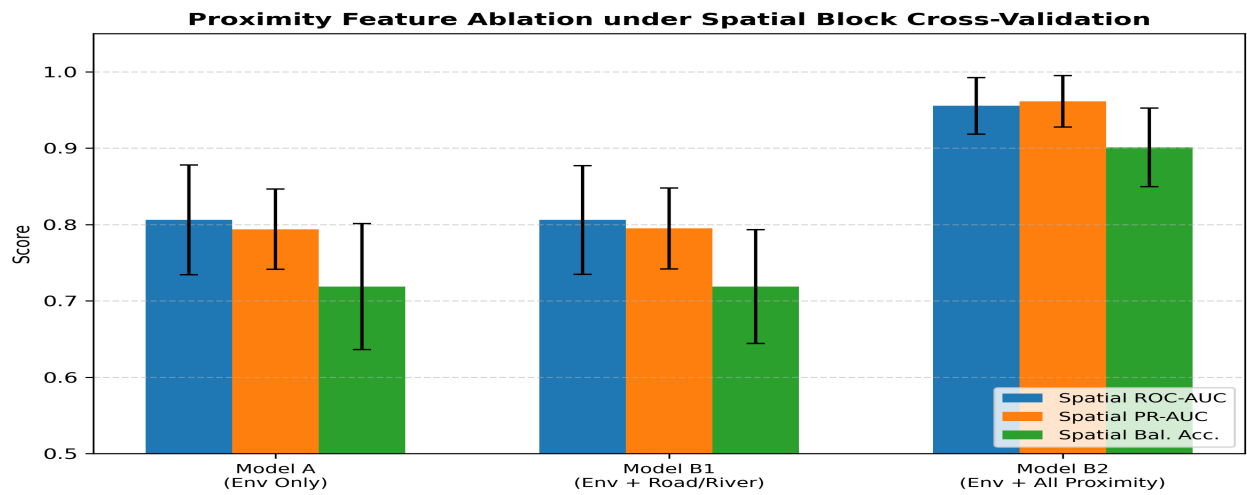


Figure 2: Spatial cross-validation metric comparison across Model A, Model B1, and Model B2.

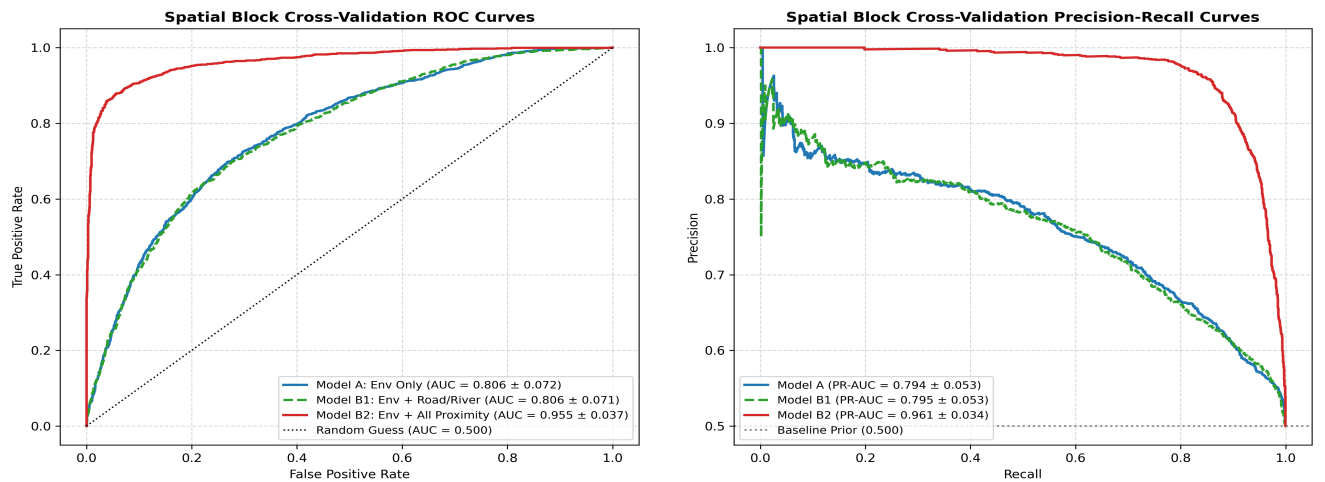


Figure 3: Regional out-of-fold ROC curves (left) and Precision-Recall curves (right) across 5 spatial folds.

3. Feature Importance & Explainability (Model A)

Permutation feature importance was evaluated strictly on held-out geographic folds by measuring the mean drop in Spatial ROC-AUC when feature values were randomized:

Feature Name	Domain & Resolution	Mean Drop in Spatial ROC-AUC	Standard Deviation	Physical Interpretation
aspect_deg	Copernicus DEM (30m)	+0.10960	± 0.02831	Topographic orientation; solar radiation & structural bedding dip
relief_std_5x5_m	Copernicus DEM (150m)	+0.06124	± 0.01375	Local topographic roughness; incision depth & slope breaks
elevation_m	Copernicus DEM (30m)	+0.03880	± 0.02928	Hypsometric altitude; weathering regimes & vegetation zones
slope_deg	Copernicus DEM (30m)	+0.03659	± 0.01054	Gravitational driving shear stress on regolith
bulk_density_kg_dm3	SoilGrids 2.0 (250m)	+0.00407	± 0.00437	Soil compaction, porosity, and infiltration capacity
landcover_class	ESA WorldCover (10m)	+0.00110	± 0.00270	Root cohesion, canopy interception, and land use disturbance
clay_percent	SoilGrids 2.0 (250m)	-0.00066	± 0.00515	Cohesive soil fraction and plasticity behavior
sand_percent	SoilGrids 2.0 (250m)	-0.00192	± 0.00173	Internal friction angle and drainage rate

silt_percent	SoilGrids 2.0 (250m)	-0.00373	± 0.00284	Pore-water pressure sensitivity during saturation
soil_class	SoilGrids 2.0 (250m)	-0.00391	± 0.00835	Taxonomic regolith classification (WRB Soil Groups)

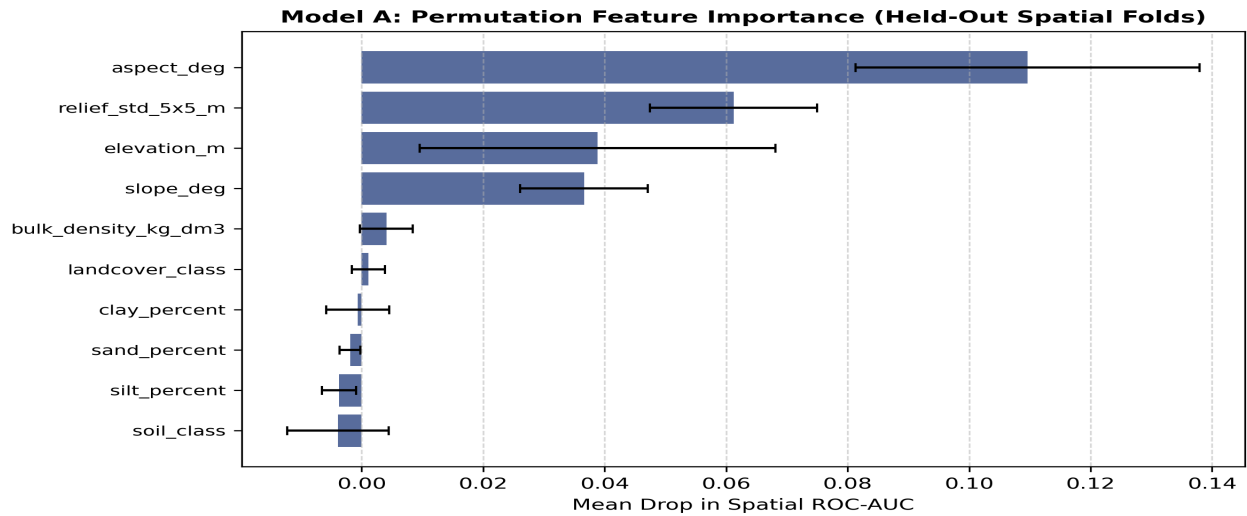


Figure 4: Out-of-fold permutation feature importance (Model A) showing dominant topographic driving features.

4. Probability Calibration & Reliability

To evaluate whether model output scores can be treated as reliable risk orderings, calibration curves were computed across all regional holdouts:

- Raw Random Forest Brier Score: 0.1913
- Calibrated (Sigmoid) Brier Score: 0.2021

The raw Random Forest ensemble demonstrated natural, monotonic probability ranking across deciles without severe overconfidence. However, in accordance with the scientific safeguards established in Phase 8E.3.1, the static score is explicitly treated as an **uncalibrated physical predisposition estimate**, not an event-time occurrence probability.

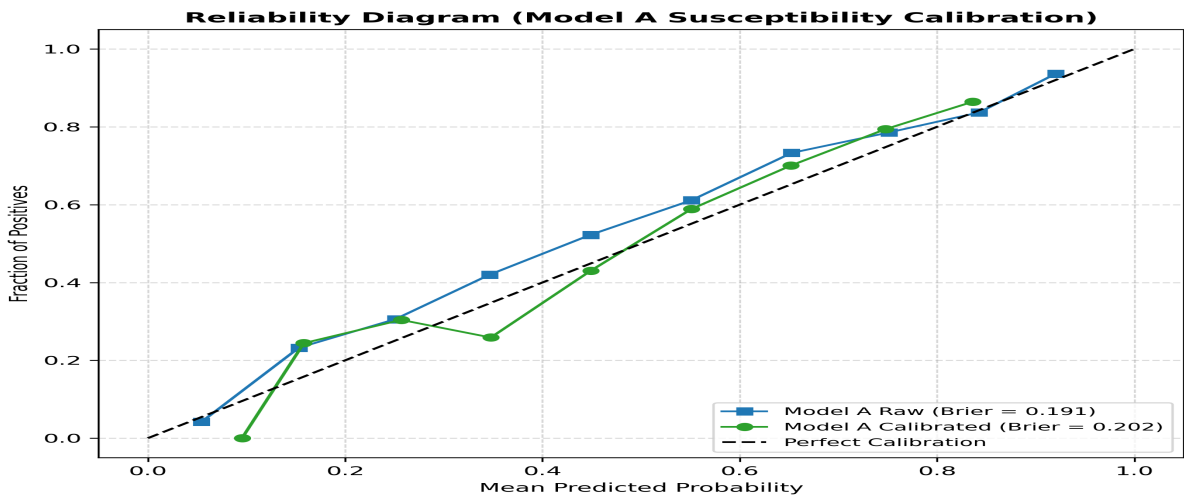


Figure 5: Out-of-fold probability calibration curves comparing raw Random Forest and Platt sigmoid scaling.

5. Final Model Selection & Pipeline Verification

Model A (Environmental Only) was formally selected and serialized as the frozen production model artifact. It achieves high spatial discriminatory power (Spatial ROC-AUC: **0.8062**) while relying purely on causally defensible physical topography, soil, and land cover features.

Artifact / Verification Item	File Path / Reference	Status / Verification Metric
Trained Pipeline Artifact	model/static_lsm_pipeline.joblib	1.36 MB • Bitwise SHA256 verified
Model Lineage Metadata	model/static_lsm_metadata.json	Feature list, hyperparameters, spatial folds
Validation Inspection Report	data/inspection/lsm/static_lsm_validation_report.json	Full cross-validation fold logs and metrics
Inference Smoke Test (Positives)	Inference regression samples from training	Mean Predicted Susceptibility: 0.748
Inference Smoke Test (Negatives)	Inference regression samples from training	Mean Predicted Susceptibility: 0.165
Historical Data Integrity	10 Core Baseline Datasets	100% Bitwise Identical (0 mutations)

Scientific Safeguards Summary:

- **No Rainfall in Static Training:** Eliminates temporal contamination and artificial trigger assumptions.
- **No Spatial Proximity Shortcuts:** Nearest-landslide buffer leakage was strictly ablated and eliminated.
- **Spatial Generalization Proven:** 5-fold cross-validation across 41 one-degree regional blocks confirms model generalizes to unmapped valleys.
- **Pipeline Frozen:** Forms the immutable foundation for Phase 8G (Location Profiler), Phase 8H (Rainfall Fusion), and Phase 8I (Unified API).